مبادئ في الإحصاء

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Memo No. 11
Date 5 / 10 / 2017

Measure of Centeral Tendency:.

(1) Mean: let $x_1, x_2, \dots, x_n$

For example $\Rightarrow$ $\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$, n = number of sample size

For population $\Rightarrow$ $\mu = \frac{\sum_{i=1}^{n} x_i}{n}$

Ex: If mean of 5 values is 8.2 and four of them: 6, 10, 7, 12

Find the fifth?

solution:. $\bar{x} = \frac{\sum_{i=1}^{5} x_i}{5} = \frac{6+10+7+12+x_5}{5} = 8.2$

$= \frac{35+x_5}{5} = 8.2$

$= 35 + x_5 = 41 \Rightarrow \boxed{x_5 = 6}$

Ex:- The mean age of 5 student is 30, anew perso whose age 25 enters class find anew mean?

solution:. $\bar{x} = \frac{\sum_{i=1}^{5} x_i}{n} \Rightarrow \sum_{i=1}^{5} x_i = n \cdot \bar{x}$

$= 5 \cdot (30) = 150$

$\sum_{i=1}^{6} x_i = 150 + 25 = 175$

$\bar{x} = \frac{175}{6} = 29.16$

Ex:. (weighted Mean)

Grads	number of student
60	5
65	4
70	1
	$n = 10$

Find Mean???

solution:. $\bar{x} = \dfrac{60 \times 5 + 65 \times 4 + 70}{10}$

Q: How to find Mean for frequency distribution of simple size?

Ex:. The age of 40 Person of Company As following find $\bar{x}$??

age	Freq (f)	midpoint (x) قيمة ممثلة	f×x
20-39	25	29.5	737.5
40-59	14	49.5	693
60-79	1	69.5	69.5
	40		1500

$\bar{x} = \dfrac{\sum f \times x}{n} = \dfrac{1500}{40} = 37.5$

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[2] Median:

Rank the data from smallest to largest:

Data: $x_1, x_2, \dots, x_{\frac{n}{2}}, x_{(\frac{n}{2})+1}, \dots, x_n$, n = # of sample size

$$\text{Median} = \begin{cases} \dfrac{x_{(\frac{n}{2})} + x_{(\frac{n}{2})+1}}{2} & : n \text{ is even} \\ x_{(\frac{n+1}{2})} & : n \text{ is odd.} \end{cases}$$

Ex: Find median for:

[1] 10, 6, 19, 8, 3, 1 :

Solution:

step(1) = $\underset{x_1}{1}, \underset{x_2}{3}, \underset{x_3}{6}, \underset{x_4}{8}, \underset{x_5}{10}, \underset{x_6}{19}$, $n=6$

$$\text{Median} = \frac{x_3 + x_4}{2} = \frac{6+8}{2} = 7$$

[2] 8, 5, 5, 4, 6, 10, 9 :

step(1) = $\underset{x_1}{4}, \underset{x_2}{5}, \underset{x_3}{5}, \underset{x_4}{6}, \underset{x_5}{8}, \underset{x_6}{9}, \underset{x_7}{10}$, $n = 7$

$$\text{Median} = x_{(\frac{7+1}{2})} = \boxed{x_4 = 6}$$

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3 Mode :.

المشاهدة القيمة الأكثر

(observation with highest frequency)

- No mode
- one mode
- two (bimodal)
- three (trmodal)
- > 3 (multi-modal)

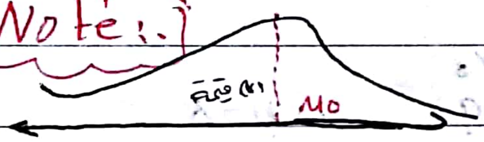
Ex:: Find Mode:.

1) 21 , 10 , 21 , 15 → 21 (one mode)

2) 60, 11, 55, 60, 70, 55 → 60, 55 (two mode)

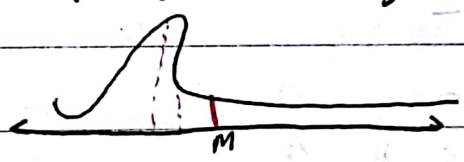
3) 60, 70, 50 → no mode .

Note:.

1) 

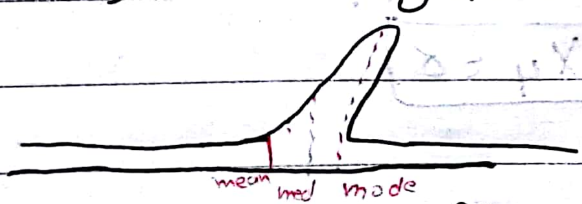
symmetrical (bell-shaped)

Mode = Mean = Median .

2) 

skew to Right

Mode < median < Mean

3) 

skew to left

Mean < Median < Mode

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Measure of dispersion or variability: مقاييس التشتت للمتغير

Range | variable | coefficient of variance (C.V)

1) Range = Largest value − smallest value

Ex: If the Range for 20, b, −10 is 40 find b???

اذا كانت $\Rightarrow b > 20 \Rightarrow \text{Range} = b - (-10) = 40$

$b + 10 = 40 \rightarrow \boxed{b = 30}$

واذا كانت $\Rightarrow b < -10 \Rightarrow \text{Range} = 20 - (b) = 40$

$\boxed{b = -20}$

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3) Veriance :

Sample

$S^2 = \sum (x_1$

Population

$$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \mu)^2}{N}$$

μ : Population mean

N : Population size.

Nots :

(1) $\sum_{i=1}^{n} x_1^2 \neq (\sum_{i=1}^{n} x_1)^2$

(2) $\sum_{i=1}^{n} (x - \bar{x}) = 0$

$\sum_{i=1}^{n} x_1 - \sum_{i=1}^{n} \bar{x}$

$\sum_{i=1}^{n} - \cancel{n} \frac{\sum x_1}{\cancel{n}}$

$= 0$

standered deviation ?

Sample: $\sqrt{S^2} = S$

Pop : $\sqrt{\sigma^2} = \sigma$

* قوانين الانحراف المعياري

Ex: Find sample vivance and sample standard deviation? 2, 3, 4, 5, 6 ? $n=5$

نعمل جدول

X_1	$X_1 - \bar{X}$	$(X-\bar{X})^2$
2	$2-4=-2$	4
3	$3-4=-1$	1
4	$4-4=0$	0
5	$5-4=1$	1
6	$6-4=2$	4
20	0	16

$\bar{X} = \frac{20}{5} = 4$

$S^2 = \sum_{i=1}^{n} \frac{(X-\bar{X})^2}{n-1}$

$= \frac{10}{4} = \frac{5}{2}$

$\sqrt{S^2} = \sqrt{2.5}$

$S = \sqrt{2.5}$

* Cofficient of variation (CV)

It's agood to compare variability of Data set with differente unit:

For sample $\Rightarrow$ $CV = \frac{S}{\bar{X}} \times 100\%$ CV قانون دل

$\bar{X}$: sample Mean (وسط حسابي)

S : Sample standard

Ex: Sample I   Sample II

$\bar{X} = 145$ $\bar{X} = 20$

$S = 10$ $S = 10$

$C.V = \frac{10}{145} \times 100\% = 69\%$

$C.V(2) = \frac{10}{20} \times 100\%$

$= 12.5\%$

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* Measures of Position:

Persentiles | Quartiles | Interquarile Rage (IQR)

percentiels:

The p-th percentile (P_i) is the value i% of the observation are less than.

Ex: $P_{100} = 70$

Q1 How to find percentiels:

step(1): arange the data (ascecondary order)

(2) $q = \left(\frac{P}{100}\right)(n+1)$ → قانون percentiels

step(3): p-th → رمز لنسبة

* في حال اذا كان عدد q نسبة أو لا يوجد قانونين

percentiles (P_i) =

(x_q), $(x_{(L)} + (q - L)[x_{(L+1)} - x_{(L)}]$ ⟹ q: is not integer

L: ineger part of (q)

له اذا كان عادي

Ex: Conseder the following sample data: (35, 12, 50, 49, 28, 42, 47)

Find: [1] The th 75 - percentile ⟹ P_{75}

يكون عدد عادي نظبق على القانون 81 ← [2] The th 40 - Percentile ⟹ P_{40}

[3] The 50^{th} - percentile ⟹ P_{50}

(Homework)

$q \left(\frac{P}{100}\right)(n+1) =$

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$\boxed{4}$ Find P_{25}, P_{30} $\Rightarrow$ Home work

(١) ترتيب تصاعديًا

(٢) نستخدم القانون لتاليا $\Leftarrow$ $q=\left(\frac{P}{100}\right)\times(n+1)$

(٣) حسب العدد الذي ينتج — عادي / غير عادي

Solution:

(1) $P_{75} = ?$

28, 35, 47, 42, 49, 50, 12

$\underset{x_1}{12}, \underset{x_2}{28}, \underset{x_3}{35}, \underset{x_4}{42}, \underset{x_5}{47}, \underset{x_6}{49}, \underset{x_7}{50}$

$n=7$

$q = \left(\frac{P}{100}\right)(n+1)$

$q = \left(\frac{75}{100}\right)(7+1)$

$q = \boxed{6} \Rightarrow (X)(q) = X(6) = 49$

عدد عادي

(2) $P(40) = ?$

$q = \left(\frac{40}{100}\right)(8)$ — * نقصد بالـ (L) الجزء الصحيح من (q)

$q = \boxed{3,2}$ — عدد غير عادي

* نستخدم القانون $(X(L) + q - L(X_{L+1} - X_L))$

$\boxed{L=3}$

$X_3 + 3,2 - 3(X_4 - X_3)$

$35 + ,2 \times 42 - 35 \Rightarrow 35 + 1.4 = \boxed{36,4}$

(3) $P(50) = ?$

$= \left(\frac{50}{100}\right)(8) = 4 \Rightarrow X_q = X_4 = 42$

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داتا
ملاحظة: $P(x)$: Median

Homework:

(4) $P_{(25)}$ = ??

$q = \left(\frac{P}{100}\right) n+1 \Rightarrow q = \left(\frac{25}{100}\right)(8) = 2 \Rightarrow X_q = X_2 = 28$

(5) $P_{(30)}$ = ??

$q = \left(\frac{P}{100}\right) n+1 \Rightarrow q = \left(\frac{30}{100}\right)(8) = \underline{\underline{2.4}} \quad L = 2$

$x_L + q - L(X_{L+1} - X_L)$

$= X_2 + 2.4 - 2(X_3 - X_2) \Rightarrow 28 + 2.4 - 2(35 - 28)$

$= 28 + 2.8 = \underline{\underline{30.8}}$

[2] Quartiles:

Q_1 = First quartile = P_{25}

Q_2 = Second quartile = P_{50} = Median

Q_3 = Third quatile = P_{75}

[3] Interquartile Range: (IQR)

$IQR = Q_3 - Q_1$

$= P_{75} - P_{25}$

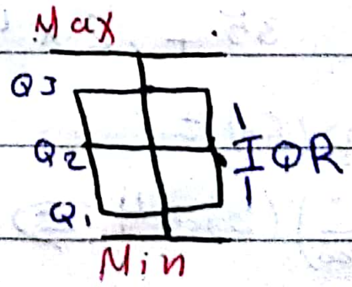
* Box-whisker plot:

Is a graphical numerical summary of five number:

Q1 Minimum value

Q2 : Median

Q3 = Max - value



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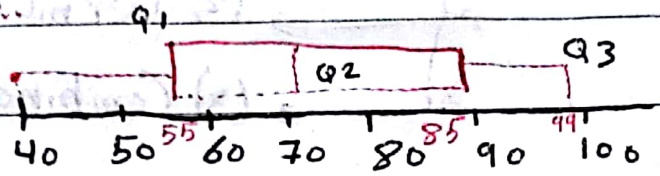
Ex:- If smallest value $= 40$

[$Q_1 = 55$, $Q_2 = 70$, $Q_3 = 85$, larget value $= 99$

Draw a box plot?

solution:-



"Normal"

* Z-Scorces:-

$$Z_i = \frac{X_i - \bar{X}}{S}$$

Note:- the mean of all Z-scorces = Zero $\bar{Z} = 0$

$S^2 = 1$ $Z : N(\overset{M}{0}, \overset{S}{1})$ (24) slide

Ex:- Consider the following Sample Data:-

12, 13, 11, 33, 34, 33, 22, 23, ?

Compute:- $\boxed{1}$ Mode $\boxed{2}$ Range $\boxed{3}$ Sample variance

$\boxed{4}$ The 90-th Percentile $\boxed{5}$ Coefficient of variation.

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ch(2): Probability :: الاحتمالات

* Counting Techniques :: → (1) multiplication Rule
  → (2) Permutations.
  → (3) Combinations.

[Count the number of ways that a phenomenon can occur.]

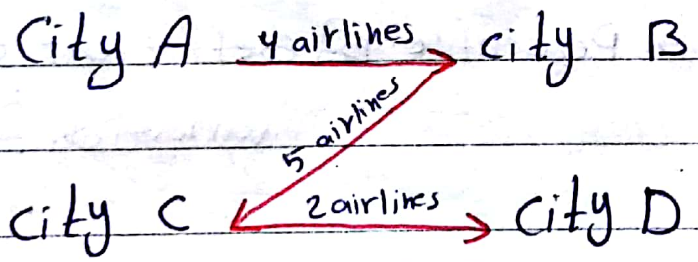
(1) muliplication Rule:

لو Experiment 1 → n outcomes احتمالات

Experiment 2 → m out comes

then the two Experiment have $n \times m$ outcomes.

EX::



How many ways from city A to city D ???

Solution::

$4 \times 5 \times 2 = 40$

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Ex (2): A perso have 3- shirts and 4 - pants.

How many different outfits?

Solution:

$4 \times 3 = 12$

Ex (3): A test consists of 5 - multiple choice questions. each question has 4- Possible Answer.

How many Possible Answers?

Solution:

$4 * 4 * 4 * 4 * 4 = 4^5 = 1024.$

Ex (4): How many Possible license Plates Could be stamped. If each license Plate were required to have exactly 3 - letters & 4 + number?

Solution:

$$\underbrace{26\ \ 26\ \ 26}\ \ \underbrace{10\ \ 10\ \ 10\ \ 10} = 26^3 * 10^4$$

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allowed → تكرار

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(2) Permutation:

A permutation is an Ordered arrangement of n different objects.

The number of Permutation of n-different element.

is $n! = n \times (n-1) \times (n-2) \times \dots \times 2 \times 1$

Recall

$4! = 4 \times 3 \times 2 \times 1 = 24$

$0! = 1$

EX: How many 3 digit-numbers can you make using the digits 1, 2, 3?

Solution: غير مسموح التكرار.

1 < 2 → 3 → 123
  3 → 2 → 132

2 < 1 → 3 → 213
  3 → 1 → 231

3 < 1 → 2 → 312
  2 → 1 → 321

$= 6 \quad (3 \times 2 \times 1 = 6 = 3!)$

اذا كان التكرار مسموح

$3 \times 3 \times 3 = 27$

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Ex:- In how many ways can 7-different books be arranged an ashelf?

Solution:-

$= \underline{7} * \underline{6} * \underline{5} * \underline{4} * \underline{3} * \underline{2} * \underline{1} = 7! = (\quad)$

Ex:- If four A, B, C and D are randomly arrenged in aline how many different arrangement such that A and B are ~~be~~ next to eachother?

Solution:- بالسؤال A, B يكونوا سوا كشخص واحد

$\underline{AB}$ ___ ___

$\underline{BA}$ ___ ___ $= (2!)(3!) = 12$

ممكن يكون اي شي

(3) Combinations:-

The number of ways in which k objects can be selected from n-different objects without regard to the order is

$$\binom{n}{k} = \frac{n!}{(n-k)!\,k!} \quad , k \leq n$$

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تزغوا أن الترتيب غير مهم

Ex: In how many ways can Pick a team of 3 - People from a group of 10 ?

Solution:

$n=10$
$k=3$

$$\binom{10}{3} = \frac{10!}{7!\,3!} = \frac{10 \times 9 \times 8 \times 7!}{7!\,3!} = 120$$

Ex: There are 5 - boys & 4 - girls in a class. A committee of 5 is to be selected such that there are 3 - boys and 2 - girls?

Solution:

$$\binom{5}{3}\binom{4}{2} = \left(\frac{5!}{2!\,3!}\right)\left(\frac{4!}{2!\,2!}\right) = 10 * 6 = 60$$

* Definitions:

(1) Random Experiment: التجربة العشوائية

An experiment than can result in different Out Comes.

(2) Sample space الفضاء العينة

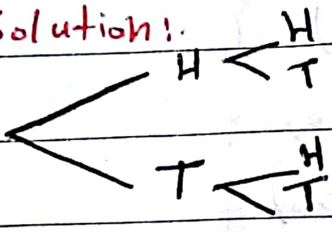
The set of all possible outcomes of a Random Experiment, denoted by S

Ex:. Find the sample space (S) for the following Random Experiment:

(1) Coin toss: $S = \{\text{Head}, \text{Tail}\} = \{H, T\}$

(2) Coins two tosses:

Solution:



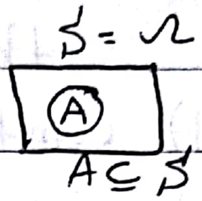
$S = \{(H,H), (H,T), (T,H), (T,T)\}$

(3) Die Roll: $\{1, 2, 3, 4, 5, 6\}$

(4) life time of an electornic compoent:

$S = \{t : t \geq 0\}$

Defn: (Event) is a subset of the sample space.
denoted by A, B, C, ...



Ex:

rolling two dice

$S = \{ (1,1), (1,2), (1,3) \ldots (1,6),$
$(2,1), (2,2), (2,3) \ldots (2,6)$
$(3,1), (3,2), (3,3) \ldots (3,6)$
$(4,1)$
$(5,1)$
$(6,1) \ldots (6,6) \}$

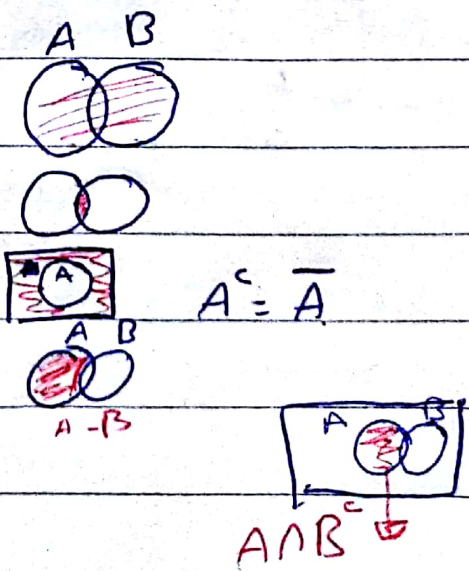
A = The sum greater than 10, 11, 12

$A = \{(5,6), (6,5), (6,6)\}$

B = The sum between 9 and 12. $\boxed{10, 11}$

$B = \{(5,5), (6,4), (4,6), (6,5), (5,6)\}$

Event operation:



1. $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
2. $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
3. $\bar{A} = A^c = \{x \mid x \in S, x \notin A\}$ $A^c = \bar{A}$
4. $A - B = \{x \mid x \in A \text{ and } x \notin B\}$
difference

$c \rightarrow$ complement

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Notes:

1. $A - B = A \cap \bar{B} \Rightarrow \{x \mid x \in A \text{ and } x \in \bar{B}\}$
2. $\bar{A} = S - A$

Rules:

1. $A \cup \bar{A} = S$
2. $A \cap \bar{A} = \{\ \} = \phi$, $A \cap S = A$
   $A \cup S = S$
3. $A \cup B = B \cup A$
3. $A \cap B = B \cap A$
4. $A \cup (B \cup C) = (A \cup B) \cup C$
   $A \cap (B \cap C) = (A \cap B) \cap C$
5. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
   $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

De Morgan's laws:

1. $\overline{(A \cup B)} = \bar{A} \cap \bar{B}$
2. $\overline{(A \cap B)} = \bar{A} \cup \bar{B}$

Defn: "classical probability)
(equally likely sample space)

If $A \subseteq S$ then the probability of A denoted by

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$$P(A) = \frac{\text{number of outcomes in } A}{\text{number of outcomes in } S} = \frac{n(A)}{n(S)}$$

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EX: Roll die. $S = \{1, 2, 3, 4, 5, 6\}$

$A =$ even number

$= \{2, 4, 6\}$

$$p(A) = \frac{n(A)}{n(S)} = \frac{3}{6} = \frac{1}{2}$$

Assumptions of ~~probability~~ probability:

1. for any event $A \Rightarrow 0 \leq p(A) \leq 1$
2. $p(S) = 1$, $p(\emptyset) = 0$
3. $p(A \cup B) = p(A) + p(B)$ iff A and B are disjoint $(A \cap B = \emptyset)$ (mutually exclusive)

* Rule of probability :-

(1) $p(A) + p(\bar{A}) = 1$

(2) If $A \subseteq B$ then $p(A) \leq p(B)$

(3) $p(A \cap \bar{B}) = p(A) - p(A \cap B)$.

(4) $p(\overbrace{A \cup B}^{A-B}) = p(A) - p(A \cap B)$

EX: If A, B are mutually exclusive. $A \cap B = \emptyset$

(disjoint) $p(A \cap B) = P(\emptyset) = 0$

such that $p(A) = 0.1$

$p(\bar{B}) = 0.4$ Find

(1) $p(B)$. $\Rightarrow p(B) = 1 - p(\bar{B}) = 1 - 0.4 = 0.6$

(2) $p(A \cap B) = 0$

(3) $P(A \cup B) = P(A) + P(B) = 0.1 + 0.6 = 0.7$

(4) $P(A - B) = P(A) - P(A \cap B) = 0.1 = 0$

(5) $P(\bar{A} \cap \bar{B}) = P(\overline{A \cup B}) = 1 - P(A \cup B) = 1 - 0.7 = 0.3$

(6) $P(\underbrace{S - A}_{\bar{A}}) = P(\bar{A}) = 0.9$

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Ex: Given A and B are (disjoint) with $P(A) = 0.2$
$P(\bar{B}) = 0.3$ Find $P(A \cup B)$? $A \cap B = \emptyset$

Solution:

$P(A \cup B) = P(A) + P(B)$
$= 0.2 + 0.7 = 0.9$

$P(B) + P(\bar{B}) = 1$
$P(B) = 1 - P(\bar{B})$
$= 1 - 0.3$
$= 0.7$

Ex: If $P(A) = 0.3$ and $P(A \cup B) = 0.7$ Find $P(B)$ If A & B are disjoint?

Solution:

$P(A \cup B) = P(A) + P(B)$
$0.7 = 0.3 + P(B$
$P(B) = 0.7 - 0.3$
$= 0.4$

Ex: The digits 1, 2, 3 are permuted randomly, what is the probability that 1 and 2 are next to each other?

Solution:

$$P(A) = \frac{n(A)}{n(S)} = \frac{(2!)(2!)}{3!} = \frac{4}{6} = \frac{2}{3}$$

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Ex:- If four people A, B, C and D are randomly arranged in aline, what is the propability that A and B are next to each other?

Solution:-

$$p(A) = \frac{n(A)}{n(s)} = \frac{(2!)(3!)}{4!}$$

2!
AB BA C D
3!

Ex:- A commitee of 5 to be chosen from agroup of 6 men and 4 women, the selection is made randomly, what is the probability that the commitee will consist of 3 men ad 2 women?

Solution:-

$$p(A) = \frac{n(A)}{SA}$$

$$= \frac{\binom{6}{3}\binom{4}{2}}{\binom{10}{5}}$$

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Conditional probability :

Defn: The conditional probility of A given B.

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad , P(B) > 0$$

Ex: A reasearch study investing the relationship between smoking and heart disease in asample of 1000 men provided the following table.

	S (smoker)	Non-S (Nonsmoker)	Total
Heart Disease	100	80	180
No Heart Disease	200	620	820
Totals	300	700	1000

(1) The probability that aselected person is smoker and has heart disease is :

$A = H \cap S =$ person is smoker and heart diease

$$P(A) = P(H \cap S) = \frac{n(H \cap S)}{n(S)} = \frac{10}{1000} = 0.1$$

(2) The probability that aselected person has aheart disease if he is smoker is :

$$P(H|S) = \frac{P(H \cap S)}{P(S)} = \frac{0.1}{0.3} = \frac{1}{3}$$

$P(S)$ خنب

$$P(S) = \frac{n(S)}{n(S)} = \frac{300}{1000} = 0.3$$

تحته عدم حامل

Two formula Involves conditional prob:

(1) multiplication Rule

(2) law of total probability

Ex: A box contains 6 red balls and 4 black balls. Two balls are drawn without replacement from the box. what is the probability that both of the balls are black?

A: Event first ball is black

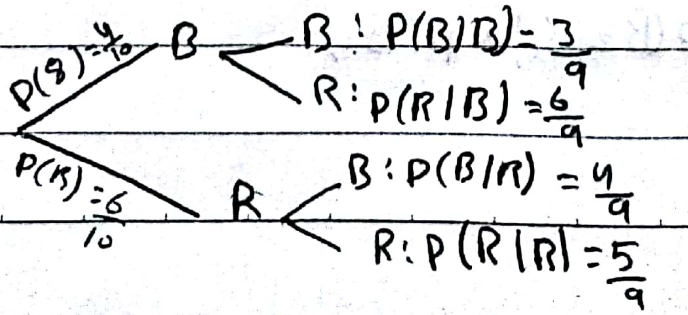
B: Event second ball is black

6-red
4-Black

solution:

$$P(A) = \frac{n(A)}{n(S)} = \frac{4}{10} \quad \Big\} \quad P(B|A) = \frac{3}{9}$$

$$P(A \cap B) = P(A|B) * P(B) \Rightarrow \frac{4}{10} * \frac{3}{9} = \frac{2}{15}$$



$$P(R \cap B) = P(B|R) + P(R)$$

red black

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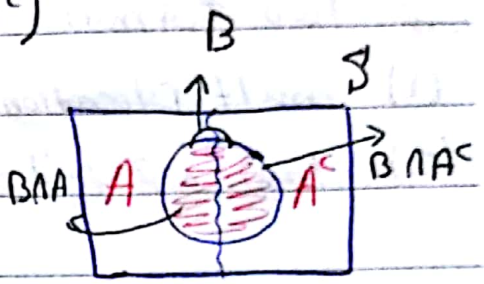
2 Law of Total Probability

Theorem: If the events A and A^c from a partition of the sample space S, then for any event B.

$$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

proof:



$B = (B \cap A) \cup (B \cap A^c)$

$P(B) = P(B \cap A) + P(B \cap A^c)$

$P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$

(1) $A \cap A^c = \emptyset$

(2) $A \cup A^c = S$

Ex: A box contains 3 blue balls and 4 red balls, 2 balls are drawn without replacement, what is the probability that the second ball is blue?

Solution:

let R_1 = event first ball is red

B_1 = event first ball is blue

B_2 = event second ball is blue. $P(B_2) = ??$

$P(B_2) = P(R_1 \cap B_2) + P(B_1 \cap B_2)$

$= P(B_2|R_1) \times P(R_1) + P(B_2|B_1) \times P(B_1)$

Mo Tu We Th Fr Sa Su

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$P(B_1) = \frac{3}{4}$

$P(R_1) = \frac{4}{7}$

$P(B_2|R_1) = \frac{3}{6}$

$P(B_2|B_1) = \frac{2}{6}$

$B(B_2) = \left(\frac{3}{6}\right)\left(\frac{4}{7}\right) + \left(\frac{2}{6}\right)\left(\frac{3}{7}\right)$

H.w

& $P(R_2)$??

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Memo No. 28
Date 19 / 10 / 2017

Q: If $p(A) = 0.7$ & $p(B) = 0.6$ and $p(A|B) = \frac{5}{6}$ Find $p(A \cup B)$??

Solution:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$
$$= 0.7 + 0.6 - 0.5$$
$$= 0.8$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$
$$P(A \cap B) = P(A|B)\,P(B)$$
$$= \frac{5}{6} * 0.6 = 0.5$$

(II) Indepedent of Events:

Defn: If $p(A|B) = p(A)$ then A & B are independent (A does not depend on B)

$*$ $$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$P(A \cap B) = P(A|B) * P(B)$ If A & B are Independent

$$\boxed{P(A \cap B) = P(A) * P(B)}$$

Result: A & B are independent $\longleftrightarrow$ $p(A \cap B) = p(A) * p(B)$

Fact: If A & B are independent then:

(1) $A, \bar{B}$ Also Indep (i.e: $P(A \cap \bar{B}) = P(A) * P(\bar{B})$)

(2) $\bar{A}, B$ Also Indep

(3) $\bar{A}, \bar{B}$ Also Indep

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Mo Tu We Th Fr Sa Su

Ex:- If $p(A) = 0.2$ & $p(B) = 0.3$ and $p(A \cup B) = 0.44$
Are A, B Indep ??

Solution:- $p(A \cap B) \overset{??}{=} p(A)\,p(B)$

$p(A \cup B) = p(A) + p(B) - p(A \cap B)$

$0.44 = 0.2 + 0.3 - p(A \cap B)$

$0.44 = 0.5 - p(A \cap B) \Rightarrow p(A \cap B) = 0.06$

للاشارة

$p(A \cap B) \overset{?}{=} p(A)\,p(B)$

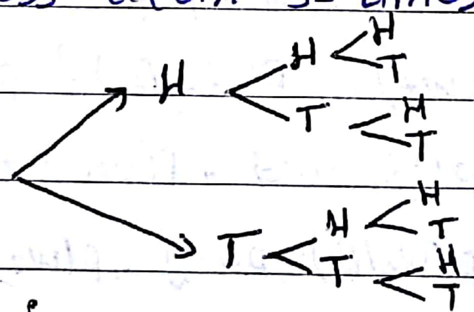
$0.06 \overset{??}{=} (0.2)(0.3) = 0.06$

Yes, Indep

ex(2) page (15):-

Toss a coin 3- times

Sol:-



$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

$\boxed{n = 8}$

let A = event of getting 2 or more heads
$= \{HHH, HHT, HTH, THH\} \Rightarrow p(A) = \frac{n(A)}{n(S)}$

$= \frac{4}{8} = \frac{1}{2}$

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B = event first toss camp up head.

$= \{HHH, HHT, HTH, HTT\}$ $p(B) = \frac{4}{8} = \frac{1}{2}$

Q: Are A & B Indep? $p(A \cap B) \overset{?}{=} p(A)p(B)$

$A \cap B = \{HHH, HHT, HTH\}$ $p(A \cap B) = \frac{n(A \cap B)}{n(s)}$

$= \frac{3}{8}$

$p(A \cap B) = p(A)\,p(B)$

$\frac{3}{8} \overset{?}{=} (\frac{1}{2})(\frac{1}{2})$ No not Indep.

Ex: (3) Page (5)

6 - blue
4 - white

Two balls are drawn (one after the other)

let B_1 = first ball is blue.

W_2 = Second ball is white.

بدون ارجاع

Q_1: Are B_1 & W_2 are Indep?

cas 1: (without replacement). $p(W_2|B_1) \overset{?}{=} p(W_2)$

$p(W_2|B_1) = \frac{4}{9}$ | $p(W_2) = p(B_1 \cap W_2) + p(W_1 \cap W_2)$

$= p(W_2|B_1) * p(B_1) + p(W_2|W_1) * p(W)$

$p(W_2|B_1) = \frac{4}{9}$

$p(W_2|W_1) = 3/9$

$p(B_1) = 6/10$

$p(W_1) = 4/10$

$p(W_2) = (\frac{4}{9})(\frac{6}{10}) + (\frac{3}{9})(\frac{4}{10})$

$p(W_2) = \frac{4}{10}$

Mo Tu We Th Fr Sa Su

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$p(w_2|B_1) \overset{??}{=} p(w_2)$

$\frac{4}{9} \overset{?}{=} \frac{4}{10}$ Not Indep.

Case 2:: (with replacement) مع الإرجاع

$p(w_2|B_1) = \frac{4}{10}$

$p(w_2) = p(w_2|B_1)\,p(B_1) + p(w_2|w_1)\,p(w_1)$

$= \frac{4}{10} * \frac{6}{10} + \frac{4}{10} * \frac{4}{10} = \frac{4}{10}$

$\frac{4}{10} = \frac{4}{10}$ Yes Indep.

Ex::

If A, B are Independent events with $p(A) = 0.2$ $p(B) = 0.5$ Find $p(A \cup B)$?

Solution::

$p(A \cup B) = p(A) + p(B) - p(A \cap B)$
$= (0.2) + (0.5) - 0.1$

because A, B Indep
$p(A \cap B) = p(A)\,p(B)$
$= (0.5)(0.2)$
$= 0.1$

$p(A \cup B) = 0.6$

Ex:: Toss Coin n- times.
what is the prob of getting all Head.

$S = \{H, T\}$

Sol::

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$A_1 =$ first event is Head $\Rightarrow p(A_1) = \frac{n(A_1)}{n(S)} = 1/2$

$A_2 =$ second " " " $\Rightarrow p(A_2) = \frac{1}{2}$

$\vdots$

$A_n =$ n-event is Head $\rightarrow p(A_n) = 1/2$

$p(A_1 \cap A_2 \cap A_3 \cdots \cap A_n) = p(A_1)p(A_2)p(A_3) \cdots p(A_n)$

by indep $= \left(\frac{1}{2}\right)^n = \frac{1}{2^n}$